Skill Builder 4A - DATA ANALYTICS - COUNT / COUNTIF / COUNTIFS

Purpose

In this activity, you will learn how to count data in Excel.

You will start by counting all records, then move to counting records that meet one condition, and finally multiple conditions.

What you will use

- COUNT
- COUNTIF
- COUNTIFS

Key Idea

We are answering one simple question:

> “How many?”

We build this in steps:

- COUNT → how many in total
- COUNTIF → how many match one rule
- COUNTIFS → how many match multiple rules

Excel will answer this by:

- looking down a column
- checking each row
- keeping a count

SECTION 1 — COUNT (no conditions)

Question 1

How many records are in the dataset?

What you do:

Use COUNT on one column

What Excel is doing:

- Excel looks down the column you selected.
- Every time it sees a value, it adds 1 to the count.
- It ignores empty cells.

So:

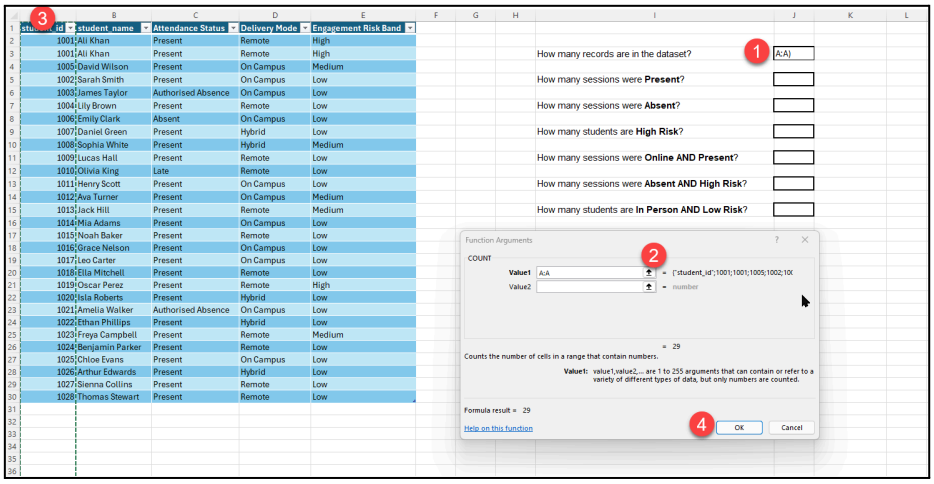
- Row 1 → count = 1
- Row 2 → count = 2
  …until the end

Final result = total number of rows with data

Important note

COUNT only works properly with numbers. If your column is text (like names), it might not count correctly.

1. Select Cell J3 and click on fx. Search for COUNT and click on Go, selecting the COUNT function front the list that is returned
2. In the COUNT Wizard, select the **Value1** box
3. Now select the **student_id** column for **Value1** (highlight all of Column A)
4. Click on **OK.** The result should be **29**



SECTION 2 — COUNTIF (one condition)

Key Idea

COUNTIF = count rows where ONE rule is true

Question 2

How many sessions were **Present**?

What Excel is doing:

1. Excel looks down the Attendance column.
2. For each row, it checks:
3. “Does this cell say ‘Present’?”

If YES → add 1
If NO → ignore it

So:

Row	Value	What happens
1	Present	Count +1
2	Absent	Ignore
3	Present	Count +1

It only counts rows that match the rule

1. Select Cell J5 and click on fx. Search for COUNTIF and click on Go, selecting the COUNTIF function front the list that is returned
2. In the COUNTIF Wizard, select the **Range** box and highlight all of Column C which contains the attendance data
3. IN **Criteria**, type **Present** (speech marks will be automatically added)
4. Click on **OK.** The result should be **25** as there are 4 entries where **Present** was not recorded (rows 6, 8, 12 and 23)

student_id	student_name	Attendance Status	Delivery Mode	Engagement Risk Band
1001	Ali Khan	Present	Remote	High
1001	Ali Khan	Present	Remote	High
1005	David Wilson	Present	On Campus	Medium
1002	Sarah Smith	Present	On Campus	Low
1003	James Taylor	Authorised Absence	On Campus	Low
1004	Lily Brown	Present	Remote	Low
1006	Emily Clark	Absent	On Campus	Low
1007	Daniel Green	Present	Hybrid	Low
1008	Sophia White	Present	Hybrid	Medium
1009	Lucas Hall	Present	Remote	Low
1010	Olivia King	Late	Remote	Low
1011	Henry Scott	Present	On Campus	Low
1012	Ava Turner	Present	On Campus	Medium
1013	Jack Hill	Present	Remote	Medium
1014	Mia Adams	Present	On Campus	Low
1015	Noah Baker	Present	Remote	Low
1016	Grace Nelson	Present	On Campus	Low
1017	Leo Carter	Present	On Campus	Low
1018	Ella Mitchell	Present	Remote	Low
1019	Oscar Perez	Present	Remote	High
1020	Isla Roberts	Present	Hybrid	Low
1021	Amelia Walker	Authorised Absence	On Campus	Low
1022	Ethan Phillips	Present	Hybrid	Low
1023	Freya Campbell	Present	Remote	Medium
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1026	Arthur Edwards	Present	Hybrid	Low
1027	Sienna Collins	Present	Remote	Low
1028	Thomas Stewart	Present	Remote	Low

How many records are in the dataset?	29
How many sessions were **Present**?	1 sent")
How many sessions were **Absent**?	
How many students are **High Risk**?	
How many sessions were **Online AND Present**?	
How many sessions were **Absent AND High Risk**?	
How many students are **In Person AND Low Risk**?	

Function Arguments

COUNTIF

2 **Range** C:C = {"Attendance Status";"Present";"Preser

3 **Criteria** "Present" = "Present"

= 25

Counts the number of cells within a range that meet the given condition.

Criteria is the condition in the form of a number, expression, or text that defines which cells will be counted.

Formula result = 25

Help on this function

4 OK Cancel

Question 3

How many sessions were **Absent**?

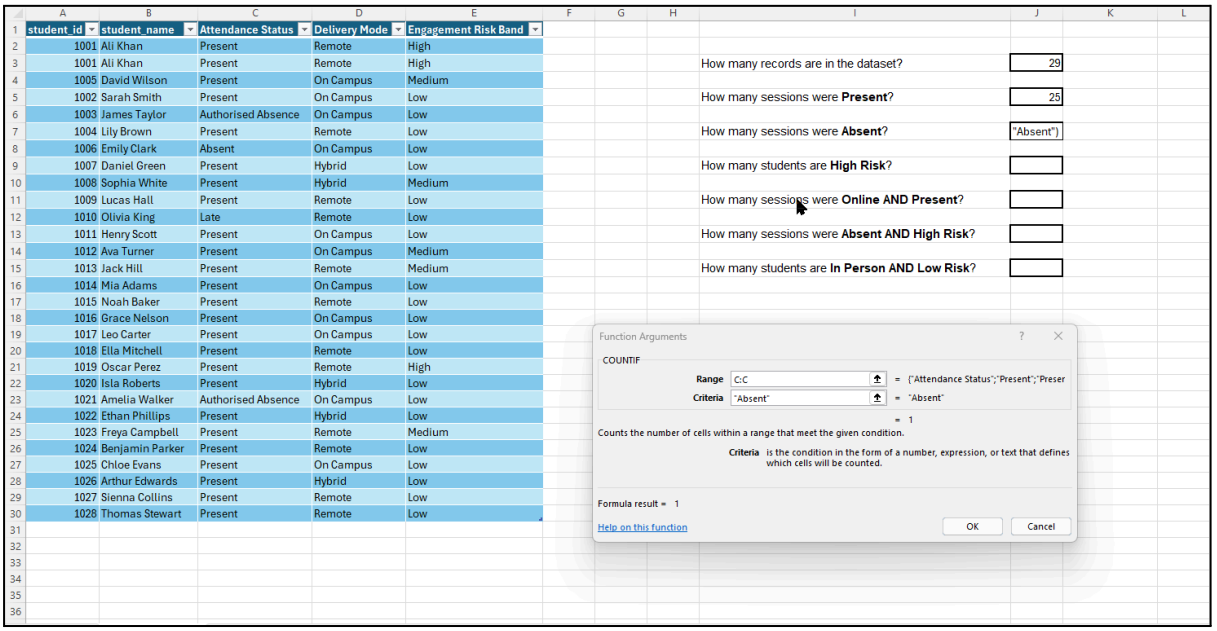
What changes?

Only the rule changes:

"Absent"

What Excel is doing:

- Same process, different rule
- It now only counts rows that say “Absent”



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How many records are in the dataset?	29
How many sessions were **Present**?	25
How many sessions were **Absent**?	"Absent")
How many students are **High Risk**?	
How many sessions were **Online AND Present**?	
How many sessions were **Absent AND High Risk**?	
How many students are **In Person AND Low Risk**?	

Question 4

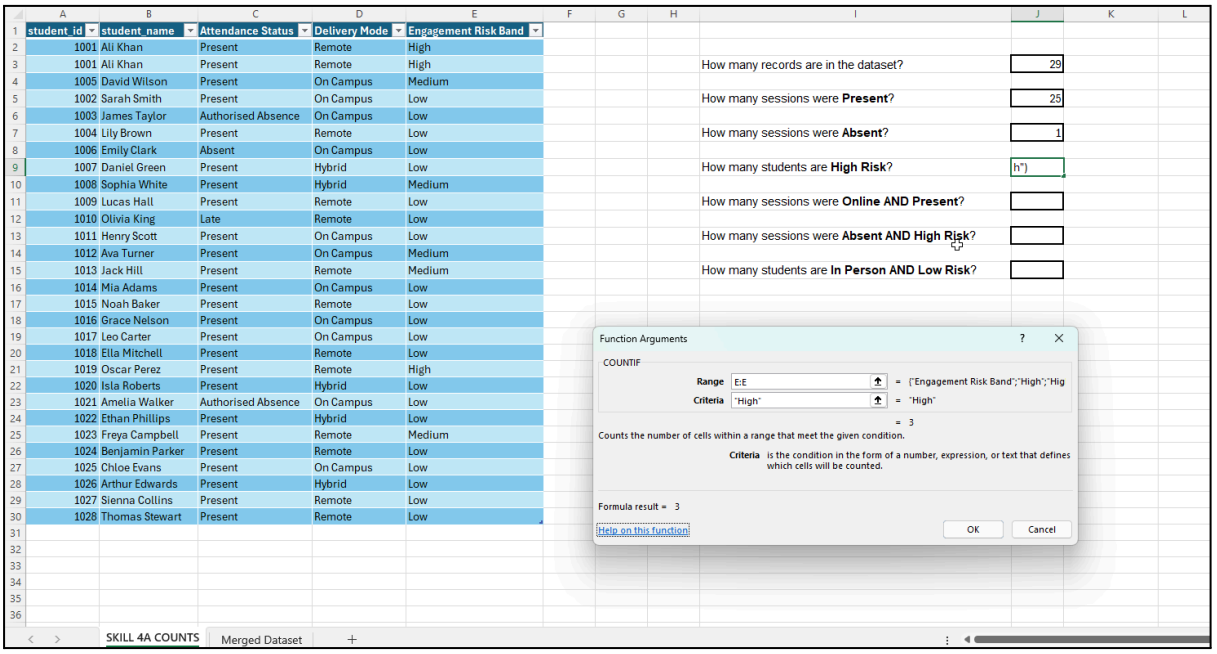
How many students are **High Risk**?

What Excel is doing:

1. Excel looks down the Risk column
2. It checks each row:
3. “Is this ‘High’?”

If yes → count it
If no → ignore it

Just be aware that you are now changing your **Range** data from *Attendance Status* to *Engagement Risk Band*



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SECTION 3 — COUNTIFS (multiple conditions)

Key Idea

COUNTIFS = count rows where MULTIPLE rules are true

> "this AND this"

Question 5

How many sessions were **Online AND Present**?

What Excel is doing:

Now Excel checks TWO things at the same time.

For each row:

1. Is Delivery Mode = Online?
2. Is Attendance = Present?

How Excel decides:

Row	Delivery	Attendance	Counted?
1	Remote	Present	✅ Yes
2	Online	Absent	❌ No
3	In Person	Present	❌ No

BOTH conditions must be true

1. Select Cell J11, click on fx and search for COUNTIFS. Click on GO to open the wizard
2. For the first data we will check - **Criteria_range1** - select *Delivery Mode* (Column D)
3. For **Critera1**, type in *Remote*
4. Press tab to go to **Criteria_range2** and select *Attendance Status* (Column C)
5. In **Criteria2**, type *Present*
6. Press OK - you should get the answer **12**

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How many records are in the dataset?	29
How many sessions were **Present**?	25
How many sessions were **Absent**?	1
How many students are **High Risk**?	3
How many sessions were **Remote AND Present**? (1)	(*)
How many sessions were **Absent AND High Risk**?	
How many students are **In Person AND Low Risk**?	

Function Arguments

COUNTIFS

(2) Criteria_range1: D:D = {"Delivery Mode";"Remote";"Remote";"

(3) Criteria1: "Remote" = "Remote"

(4) Criteria_range2: C:C = {"Attendance Status";"Present";"Preser

(5) Criteria2: "Present" = "Present"

Criteria_range3: = reference

= 12

Counts the number of cells specified by a given set of conditions or criteria.

Criteria_range3: is the range of cells you want evaluated for the particular condition.

Formula result = 12

Help on this function

(6) OK Cancel

Question 6

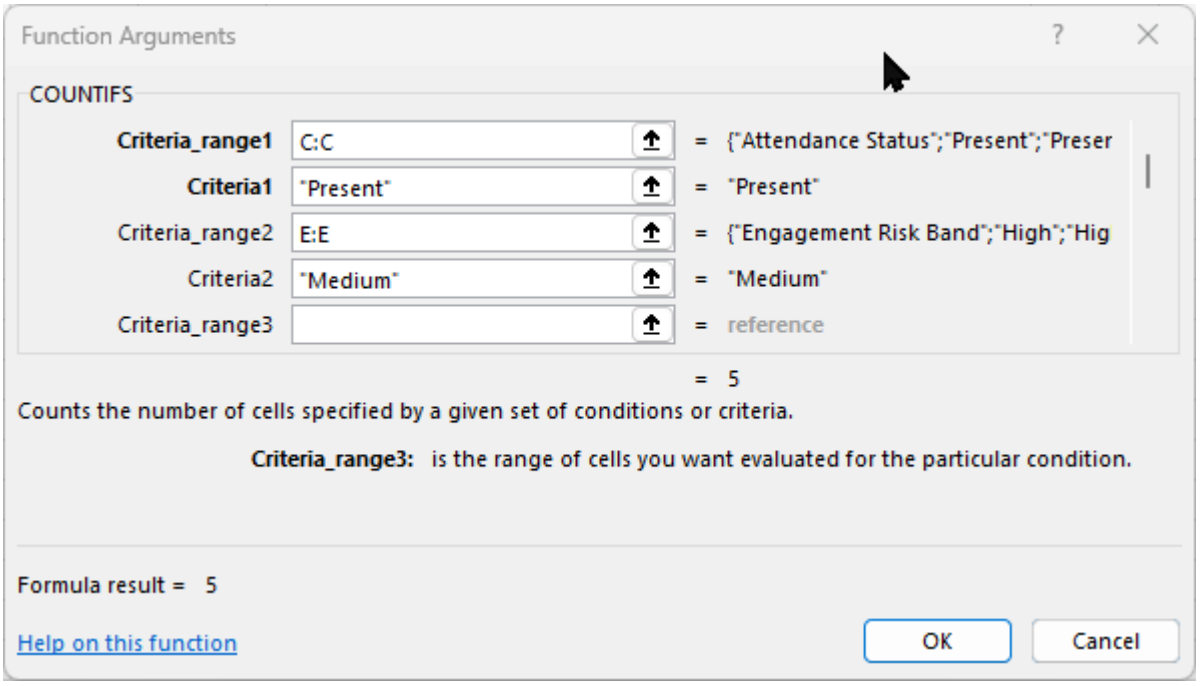
How many sessions were **Absent AND High Risk**?

What Excel is doing:

For each row:

- Is Attendance = Absent?
- AND is Risk = High?

Only count if BOTH are true



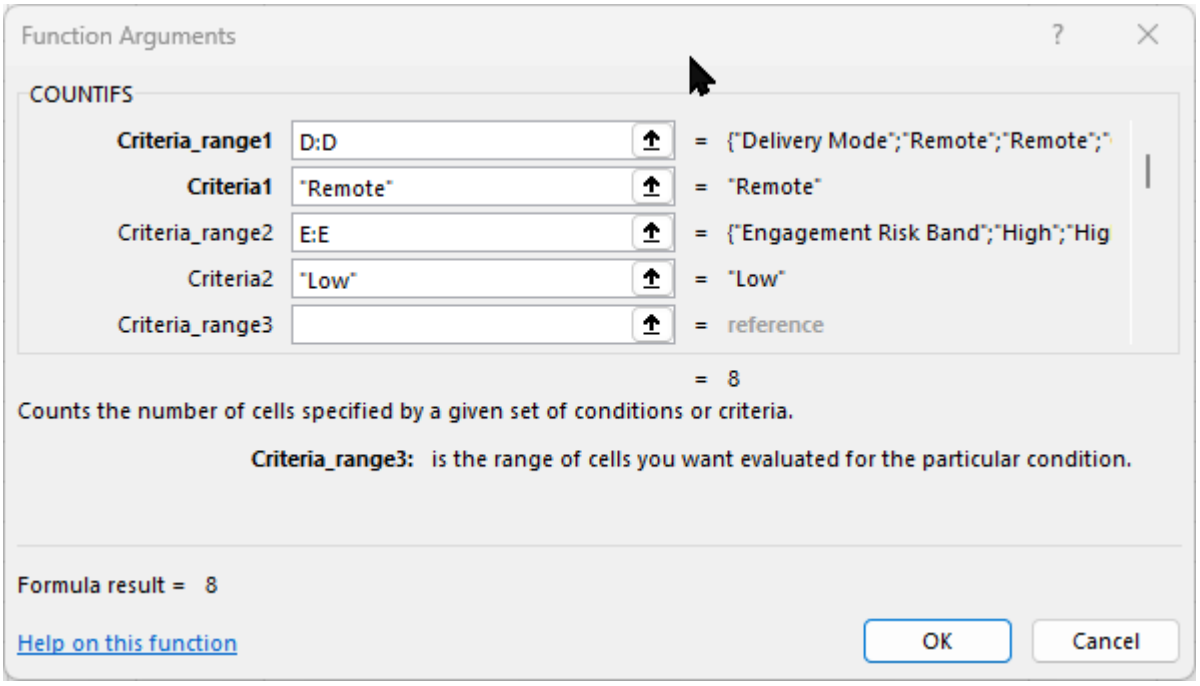
Your answer should be **5**

Question 7

How many students are **Remote AND Low Risk**?

Same logic again:

- Excel checks each row
- It only counts rows where BOTH conditions match



Final Summary

What Excel is doing every time:

- It looks at one row at a time
- It checks your rule(s)
- It decides: count it or ignore it

The only difference between functions:

- COUNT → count everything
- COUNTIF → count if ONE rule is true
- COUNTIFS → count if MULTIPLE rules are true

The simplest way to remember:

- COUNT = “How many?”
- COUNTIF = “How many match this?”
- COUNTIFS = “How many match this AND this?”

Final note

If you get stuck, come back to this:

> “Excel is just going row by row and checking rules”